Completing Missing Productive Rejections for Boundary Decisions

Experiment index

Question and Context

Did the first preference set underteach "clarify instead of act" and "refuse instead of act" because those rejected alternatives were missing for many anchors?

Change

Start fresh from the original anchored two-epoch SFT adapter, not the previous DPO adapter or the 2:1 SFT model.

Add 95 explicit incorrect ACT answers to existing training prompts: 50 clarification and 45 refusal anchors. Negatives deliberately guess missing details, skip confirmation, call a known unavailable operation, or substitute an inappropriate productive action.

Training-only counterparts take priority. Authoring explanations stay in the review sidecar, not the model input. No evaluation fixture supplied a target.

Preference Artifact

Chosen over rejected	Before	After
ACT > CLARIFY	200	200
ACT > REFUSE	200	200
CLARIFY > ACT	50	100
CLARIFY > REFUSE	100	100
REFUSE > ACT	55	100
REFUSE > CLARIFY	100	100

800 pairs, still 400 unique prompts and the same 200/100/100 unique anchor counts. Actual artifact: 25,854 chosen and 24,541 rejected completion tokens, longest sequence 633, no truncated pairs.

- Start policy and frozen reference from the specified SFT adapter, not bare Qwen.
- One DPO epoch, sigmoid preference loss, beta 0.1, learning rate 5e-6, constant scheduler.
- Batch one, accumulation 16, context 1,024, seed 1; inherited LoRA rank/targets.
- TRL 1.14.0 ordinary PyTorch full-logits path, with reference log probabilities precomputed.

- Chosen/rejected answers share the same prompt; no JSON repair or relaxed evaluator.
- Rule-derived training preferences plus owner review, not LLM-judge labels.
- No DPO validation preference split was constructed; evaluation used the existing development suite.

Recorded Preference Artifact Summary

Artifact measurement	Recorded value
Preference pairs	800
Unique training prompts	400
Chosen completion tokens	25,854
Rejected completion tokens	24,541
Longest complete sequence (tokens)	633
Truncated pairs	0

Preferred decision	Pair exposure
ACT	400
CLARIFY	200
REFUSE	200

Pair exposure counts comparisons, not unique anchors. One prompt can contribute multiple distinct rejections.

Source: recorded DPO run. This export-only table transcribes recorded metadata; it does not change the source note or any training artifact.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
DPO / completed ACT rejections	150	46/150 (0.307)	0.533	0.793	0.050	0.155

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
DPO / completed ACT rejections (150 cases)	0.482	0.047

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: DPO / completed ACT rejections (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	60	14	1	6
CLARIFY	30	9	1	0
REFUSE	15	2	11	1

Result Provenance

- DPO / completed ACT rejections: 2026-10-09 UTC, 150 cases.

Interpretation

Task successes were 46/150, macro-F1 0.482 and invalid outputs seven. Compared with first DPO's 53 successes / 0.376 macro-F1, this recovered some routing balance at a task-completion cost.

Premature ACT decisions fell from 56 to 45, but the original SFT reference had only 31. Original SFT still had higher macro-F1 (0.514) and fewer invalid outputs (two).

The gap repair helped relative to the first DPO candidate, but did not beat the original SFT on balanced routing.

Limits

Adding 95 pairs increases optimizer exposure at the same epoch count: this is not a step-matched ablation. Coverage is complete but directions are not equally weighted: ACT-over-boundary has 400 pairs, boundary-over-ACT 200.

No claim follows that DPO cannot work, that Qwen is inherently unsuitable, or that longer DPO would fix it. Those remain untested hypotheses.

Supporting Files

- qwen2_5_1_5b_anchored_epochs2_act_rejections_v2.json
- dpo-run.json
- dpo_act_rejections.py

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.